International Journal of Green Energy

ISSN: 1543-5075 (Print) 1543-5083 (Online) Journal homepage: https://www.tandfonline.com/loi/ljge20

The potential and economic viability of wind farms in Zimbabwe

Remember Samu, Murat Fahrioglu & Cagil Ozansoy

INTERNATIONAL JOURNAL OF GREEN ENERGY
https://doi.org/10.1080/15435075.2019.1671424

The potential and economic viability of wind farms in Zimbabwe

Remember Samu [a], Murat Fahrioglu [b], and Cagil Ozansoy [c]

[a]Discipline of Engineering and Energy College of Science, Health, Engineering and Education, Murdoch University, Murdoch, Australia; [b]Electrical and Electronics Engineering Department, Middle East Technical University Northern, Mersin, Turkey; [c]College of Engineering and Science, Victoria University, Melbourne, Australia

ABSTRACT

Wind energy potential in Zimbabwe so as to curb for the energy deficit is assessed in this study. The frequency of load shedding in Zimbabwe has increased; this is due to insufficient energy generation and rising energy demand. Wind energy is intermittent and these fluctuations might lead to an unreliable system so the wind turbines are going to be grid-connected, in the event of a deficit from the wind system, the grid will supply. Conversion of wind resources in 28 different locations scattered all over Zimbabwe into electrical energy is the main objective of this study. The study shows the energy production cost ranges from US$77.23/MWh to US$129.46/MWh. This kind of range will allow wind energy to be used in the future energy plans of Zimbabwe. Integration of renewable energy resources to increase the generation capacity does not only increase the energy production but also ensures a cleaner energy generation mix which is environmentally friendly.

ARTICLE HISTORY
Received 12 July 2018
Accepted 15 September 2019

KEYWORDS
Environmental effects; grid-connected; renewable energy; RETScreen modeling software; wind energy; Zimbabwe

1. Introduction

Depletion of fossil fuel resources, global warming, climate change and fluctuations in oil prices have led to an urgent need for cleaner energy generation methods. These are not only environmentally friendly but practically economic once the externalities are internalized, that is the inclusion of other factors like the social cost of carbon into the financial analyzes (Samu and Fahrioglu 2017). Regardless of its intermittency, high initial cost and vast land requirements, wind energy is one of the fast-growing clean energy technology worldwide.

Studies concerning issues related to wind energy harnessing and the factors that influence wind farms' performances have been performed worldwide. The goals that most researchers try to achieve are; optimal sight choosing, maximization of wind power generation, maximizing the power factor and minimizing the cost of electricity (Cetinay, Kuipers, and Nezih Guven 2017; Genchi et al. 2016; Gualtieri 2019; Gualtieri and Zappitelli 2015; Santos and Mario 2019). Due to the vast wind offshore and onshore wind resources in Turkey other many other parts of Europe, an enormous amount of wind potential studies and site choosing optimization studies have been performed in these regions. Offshore studies include but not limited to; an analysis performed for the Black Sea Region by (Argin and Yerci 2017), in which they concluded that for the optimal location criteria, more parameters other than just wind speeds should be considered. Studies performed in Hatay, Manisa and other parts of Turkey, proved the feasibility of wind farm constructions in all the regions (Cetinay, Kuipers, and Nezih Guven 2017; Ersoz et al. 2013; Sahin and Bilgili 2009; Ucar and Balo 2008). Statistical analyses for the wind energy potential analysis were done by (Commin and John 2017; Gugliani et al. 2019, 2017; Hedayat et al. 2017; Türk Toğrul and Kizi 2008; Xydis 2016), in which they proved the effectiveness of distributions like the Weibull and the Rayleigh for wind potential studies. On top on these distributions, a study in Malaysia by (Sanusi et al. 2017), added a von Mises distribution to the analysis of wind energy potential, in which they concluded that although many studies consider wind speeds only, dominant wind direction is also of importance.

Vast clean energy resources are available in the whole of Africa, but they are not being utilized as well. This leads to not only a deficit in energy supplies but also to the utilization of unclean fossil fuel resources. Sub-Saharan Africa contributes 13% of the world population today, but only 4% out of this 13% have access to basic energy. On the bright side, it has been estimated that there has been a rapid growth in energy accessibility by 45% since 2000 (International Energy Agency 2014). Studies on the wind potential of Sub-Saharan Africa estimates a total potential of 1,300 GW which is way above the current demand of the whole of Africa (Mandelli, Barbieri, Mattarolo, and Colombo 2014; Samuel Asumadu- Sarkodie 2016).

Currently, the total installed generation capacity in Zimbabwe is 1,960 MW against a peak demand of 2,200 MW. Of the 1,960 installed, only 1,605 MW are operational (Zimbabwe Electricity and Transmission Distribution Company (ZETDC) 2017), thus leaving a 35% deficit in energy supply. The installed fossil fuel capacity is 1,220 MW and the hydro capacity is 750 MW. In terms of percentages, Coal constitutes 62% and the rest 38% is Hydro (Africa-EU Renewable Energy Cooperation Program 2015).

CONTACT Cagil Ozansoy cagil.ozansoy@vu.edu.au Electrical, Electronic and Sports Engineering Department, College of Engineering and Science, Victoria University, PO Box 14428, Melbourne, Victoria, Australia

Color versions of one or more of the figures in the article can be found online at www.tandfonline.com/ljge.

Renewable energy resources utilization is still lagging behind in Zimbabwe. Plans to invest in solar PV technology are at an advanced stage as Zimbabwe is looking forward to the installation of 300 MW (Samu and Fahrioglu 2017). Zimbabwe as a landlocked country is not rich in wind energy resources (Samu, Fahrioglu, and Taylan 2016; Tawanda Hove, Luxmore Madiye 2014). However, with the advances in technology leading to the development of wind turbine prototypes with a lower cut in speeds, it can be possible to harness wind energy in Zimbabwe. Wind energy had been used for pumping water to meet farm requirements using windmills. A study on the wind potential in Zimbabwe estimated that it might be possible to generate electricity at a hub height of 80 m, resulting in the manufacturing of wind turbine prototypes rated at 4 kW and 1 kW, which tested well and were installed in Rusape (World Bank 2016; ZERA Farmers INDABA Workshop 2015 2015).

However, there is little scientific research on wind energy potential in Zimbabwe. With wind speeds around 3.2 m/s, most parts of Zimbabwe have wind potential only viable for pumping purposes. The Eastern Highlands and some parts of Bulawayo have with higher wind speeds (Mungwena 2002). Another study in Gwanda, located in the Matabeleland province of Zimbabwe, confirmed a yearly average wind speed of 4.75 m/s (Samu, Fahrioglu, and Taylan 2016). With this average wind speed, it is, therefore, possible to harness wind energy utilizing wind turbines with low cut-in speeds since it was reported by (Yumak, Ucar, and Yayla 2012) that any average wind speeds above 3 m/s are acceptable for wind energy harnessing. (Yumak, Ucar, and Yayla 2012) came to this conclusion after their study in Mamedik, whilst (Manomaiphiboon et al. 2017), also proved that turbines with a low cut-in wind speed were also the best for Thailand.

Further related studies for wind potentials confirmed the feasibility of high yearly yields of wind energy in many African countries (Mentis et al. 2015). (Adaramola, Agelin-Chaab, and Paul 2014) did a wind energy analysis along the coasts of Ghana in which they only took into account six stations. By employing different methodologies, wind potential assessments were carried out in many different parts of Africa (Boudia et al. 2012; Himri, Boudghene Stambouli, and Draoui 2009; Landry et al. 2017). Southern African potential analyses were performed for countries such as South Africa and the Island of Mauritius (Ayodele et al. 2014, 2013; Fant, Gunturu, and Schlosser 2016; Oree and Rajoo 2015), but none had been performed for Zimbabwe.

The contribution of this present study is to analyze the economic viability and potential of wind energy in Zimbabwe employing RETScreen modeling software. As the energy situation in Zimbabwe is deteriorating, this analysis will highlight the feasibility of how the country can help solve the current energy shortages in an environmentally friendly as well as sustainable manner. There has not been any analysis of the wind potential for Zimbabwe and against this backdrop, this present study analyses the wind energy potential in Zimbabwe in terms of; energy production costs and savings, energy production, operation and maintenance costs, life-cycle costs, reduction in greenhouse gas emissions and financial feasibility of 10 MW wind farms for 28 different stations scattered all over Zimbabwe. These 28 locations are scattered across the whole country and were chosen due to the availability of wind resources, availability of land, road network infrastructure and accessibility to the national grid for connection. Some of Zimbabwe's vision 2030 goals include ensuring a greenhouse gas emissions reduction of 33% by 2030, drafting a renewable energy policy targeting harnessing 1000 MW of renewable energy by 2025 and ensuring sustainable energy for all (Aecf 2018; Makonese 2016). To the knowledge of the authors up to date, there is no renewable energy policy in force in Zimbabwe and a study by (Samu, Fahrioglu, and Taylan 2016), suggested that it might be feasible to harness wind energy in Zimbabwe if low-cut-in speed wind turbines are utilized. Another study by (Remember Samu and Fahrioglu 2017) suggested that almost 300 MW can be harnessed from solar PV in Zimbabwe, with the same goal to help policymakers as well as all stakeholders to consider wind energy harnessing as a possibility in Zimbabwe, an analysis is, therefore, performed in these 28 locations in this present study, which will give a total of 280 MW.

2. Methodology

Meteonorm V7 software was used to generate Typical Meteorological Year 2 (TMY2) wind speed data for the 28 locations in Zimbabwe. RETScreen modeling software was then employed to further analyze the wind energy potential. Typical Meteorological Year (TMY) data would have been more accurate for the present analysis because of the atmospheric losses due to changes in meteorological conditions (NREL.gov 2015). However, it is difficult to access TMY data, and hence TMY2 data developed by NREL was instead used.

In this study, the mathematical details of the generation of TMY2 data were excluded due to their availability in the already existing literature. The RETScreen software used in this study was developed by National Resource Canada. An assessment of the energy production costs and savings, energy production, operation and maintenance costs, life-cycle costs, reduction in greenhouse gas emissions and financial feasibility determination of the wind energy systems were done using the RETScreen modeling software. Figure 1 is an operational model framework of the RETScreen modeling software and detailed theory of the energy calculations is embedded in the software. This excel-based software has multiple embedded models which include the financial analysis model (FAM), the energy model, the sensitivity and risk analysis model (SRAM) and the greenhouse gas emission reduction analysis model. The Monte Carlo simulation, risk analysis model validation, the median and confidence interval and the impact graph are included in the SRAM, whilst the debt payments, income tax, asset depreciation, financial feasibility indicators and the pre-tax and after-tax cash flows are included in the FAM.

The locations under study were chosen based on their accessibility to the grid, availability of their data set from Meteonorm V7 software and NASA global solar radiation database. The geographical locations of the locations as well as the respective average wind speeds are shown in Table 1. For the accurate analysis of wind potential, there is a need to take into account some losses and full consideration of some important coefficients. The wind turbine used for the analysis in this study is

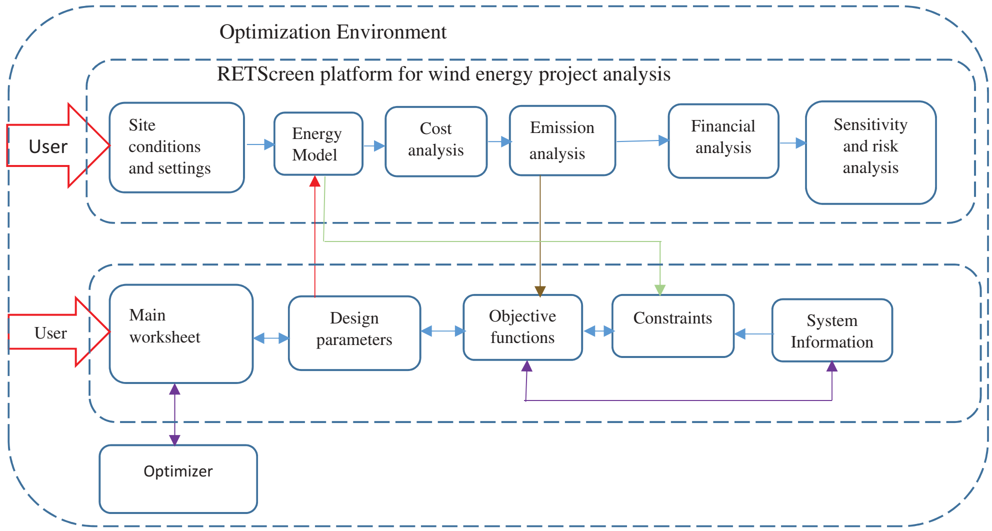

Figure 1. An operational model framework of the RETScreen modeling software.

Table 1. Coordinates and average wind velocities of the 28 locations in Zimbabwe.

Location	Longitude (°E)	Latitude (°N)	Elevation (m)	Annual Average Wind Velocity (m/s)
Beitbridge	30.00	22.20	696	3.4
Bulawayo	28.60	20.20	1,344	3.2
Chegutu	30.10	18.10	1,261	3.4
Chikore	32.30	17.80	974	3.6
Chinhoyi	30.20	17.40	1,293	3.5
Chipinge	32.60	20.20	576	3.6
Chiredzi	31.70	21.10	425	3.5
Dorowa	31.80	19.10	1,106	3.5
Gokwe	28.90	18.20	1,078	3.5
Gwanda	29.00	20.90	1,096	3.2
Gweru	29.80	19.50	1,311	3.3
Gwetera	32.00	16.90	564	3.6
Harare	31.00	17.80	1,472	3.6
Kutsaga	31.10	17.90	1,480	3.5
Hwange	26.50	18.40	905	3.6
Inyati	28.90	19.70	1,192	3.3
Kamativi	27.10	18.30	967	3.6
Kariba	28.80	16.50	646	3.4
Karoi	29.70	16.80	939	3.5
Kwekwe	29.80	18.90	1095	3.4
Marawa	29.30	17.80	960	3.5
Marondera	31.60	18.20	1,392	3.6
Masvingo	30.90	20.10	1,095	3.2
Mhangura	30.20	16.90	862	3.5
Mt Darwin	31.60	16.80	782	3.6
Mutare	32.70	19.00	1,332	3.7
Ruwa	32.60	19.20	976	3.7
Shurugwi	30.00	19.70	1,281	3.4

Gamesa G132-5.0 prototype and its specifications together with related coefficients and wind energy losses are given in Table 2.

2.1. Financial analysis

An analysis of the wind energy potential is of importance to the investors and decision-makers in both the private and public sectors. RETScreen modeling software version was employed in this present study for the financial feasibility study of proposed 10 MW wind power plants. Some of the financial parameters used to perform this analysis are; energy cost, inflation rate, greenhouse gas emission credit, project lifetime, electricity export escalation rate, etc.

Table 2. Wind turbine specifications, wind energy losses and its related coefficients.

Variable	Corresponding Value
Turbine Rated power	5,000 kW
Turbine cut-in wind speed	1.5 m/s
Turbine rated wind speed	13 m/s
Turbine cut out speed	27 m/s
Turbine Hub height	95 m
Rotor diameter	132 m
Swept area	13,685 m^2
Wind shear constant	0.14
Array losses	7%
Airfoil soiling and/or icing losses	3%
Miscellaneous losses	5%
Downtime losses	10%
Availability	98%
Temperature adjustment coefficient	0.96

The financial input parameters and some assumptions in Table 3 were used for the cost analysis. The source of some input parameters is RETScreen Clean Energy Project Analysis software unless otherwise stated.

In the first phase, before the construction of the wind farm, the following were considered:

- A feasibility analysis comprising resource assessment, detailed cost estimate, site investigation, environmental assessment, greenhouse gases baseline study, project management, report preparation, and travel and accommodation.
- The development, consisting of site survey and land rights, project financing, contract negotiation, project management, permits and approval, travel and accommodation, greenhouse gases validation and registration, and legal and accounting.

During the implementation of the wind power project, engineering, power system and balance of system and miscellaneous were considered. After project implementation, the periodic costs, contingencies and operation and maintenance costs were then considered. From the analysis, in total, the investment of a 10 MW wind farm costs US$17,085,300 and

Table 3. Input parameters for the financial analysis (OpenEI 2018).

Parameter	Value
Energy cost escalation rate	5%
Inflation rate	−0.95% (Zimbabwe Inflation Rate 2016)
Discount rate	7.17% (Zimbabwe Central Bank Discount Rate 2016)
Interest rate	10.71% (Zimbabwe Interest Rate 2016)
Project lifetime	20 years (NREL.gov 2015)
Debt ratio	70%
Debt interest rate	7%
Debt term	20 years
Transmission and distribution losses	18% (International Energy Agency 2014. World Energy Outlook Special Report. In Africa Energy Outlook 2014)
Electricity export escalation rate	4%
Development	US$63/kW (NREL.gov 2015)
Installed Capital Cost	US$1,200/kW
Feasibility study	US$35,300
Balance of plant	US$418/kW (NREL.gov 2015)
Renewable energy production credit	0.025 $/kWh
Miscellaneous	US$418/kW (NREL.gov 2015)
Annual Operation costs	US$10/kW/year
Engineering	US$24/kW (NREL.gov 2015)
Renewable energy production credit duration	10 years
Renewable energy credit escalation rate	2.5%
GHG emission reduction credit	40 $/$t_{CO2}$ (EPA 2016)
GHG reduction credit	21 years
Energy cost escalation rate	5%

using the rule of thumb by Tom Gray, 60 acres per megawatt that are 242,811 m2/MW is required for the proposed project (Aweo.org 2015). The flow chart in Figure 2 is the simplified operational summary of the grid-connected wind energy system, where, E_{gen} is the energy generated by the renewable energy system (RES), which is the wind energy generation system in the case of this present study, D is the electrical energy demand of the location, (D_{RES}) is the demand met by the RES, (D_{grid}) is the demand provided by the grid and (D_{excess}) is the excess energy generated by the RES, in this study this (D_{excess}) is fed to the grid, free of charge.

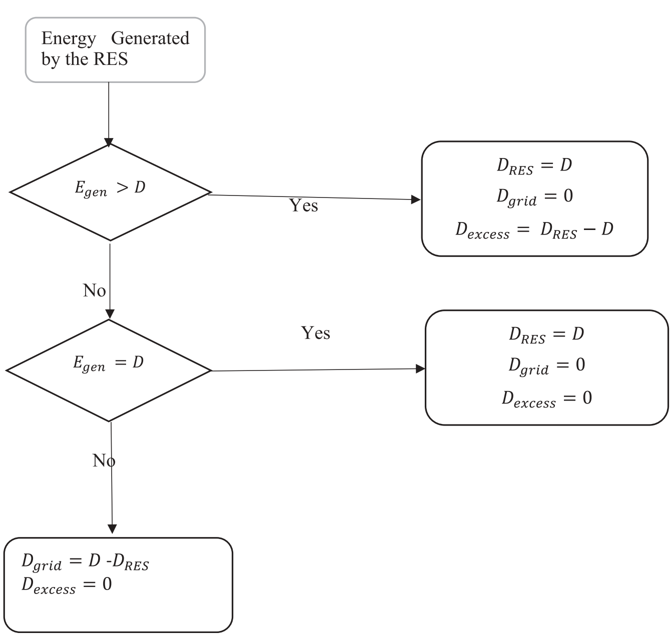

Figure 2. The flow chart of the wind energy generation system.

3. Results and discussion

For the analysis of wind energy potential in Zimbabwe, Gamesa G132-5.0 MW wind turbine was used. This wind turbine was selected because it is the most powerful wind turbine (Gamesa 2016). This wind turbine model also has a low cut-in speed of 1.5 m/s (Gamesa 2016), thus maximizing energy generation at low wind speeds making it most suitable for Zimbabwe which has low average wind speed (Samu, Fahrioglu, and Taylan 2016). The electricity export rate in this present study was taken as US$200/MWh (Henbest, Mills, Orlandi, Serhal and Pathania 2015).

For the determination of feasible locations, a few noteworthy parameters were used. These include the gross energy production, specific yield, electricity exported to the grid, capacity factor, net present value, simple payback, the energy production cost, net annual GHG emission reduction and resulting avoided crude oil consumption in the 28 locations. A summary of the 10 MW wind farms in 28 different locations is shown in Table 4. As the results in Figure 3 show, the higher the capacity factor present at a certain location, the higher the output gross energy production. Figure 4 is clearer and pictorial summary and proof of the relationships between the parameters under analysis for the wind energy potential in Zimbabwe.

The results outline the gross energy production of the Gamesa G132-5.0 MW wind turbine with a rated power of 5000 kW. It is evident from the results that all the locations under study produce energy above 4000 MWh. Contrary to some previous studies by (News 2015; Samu, Fahrioglu, and Taylan 2016; Tawanda Hove, Luxmore Madiye 2014), the results of this study prove otherwise because the employed wind turbine model has a relatively low cut-in speed and low rated wind speed thus favoring all the sites under study.

In Table 4, column 6, the specific yield of the wind turbine is reported. This is a more accurate parameter used for the calculation of wind energy cost ($/kWh) since it takes into account the annual energy output per square meter of the swept area by the wind turbine blades upon rotation (Windenergyfoundation.org 2015). Since this specific yield is directly proportional to the capacity factor as well as the wind speed, all locations under study produced an annual energy above 250 kWh/m2, though Ruwa had the highest yield of 409 kWh/m2 followed by Chipinge with 408 kWh/m2 and the lowest was 275 kWh/m2 observed in Bulawayo.

The investment financial feasibility is highly depended on the NPV. If the NPV is positive the project is potentially feasibility if otherwise, then the project is rejected. A maximum NPV of 20.3 million$ is observed at Ruwa location while the minimum value is 7.8 million$ observed at Bulawayo. The simple payback period is reported in Table 4 column 3. The shortest simple payback period is 6.3 years for Ruwa whilst the longest is 9.6 years for Bulawayo. The longer the simple payback, the less desirable is the investment (Himri, Boudghene Stambouli, and Draoui 2009). However, the project lifetime is 20 years, therefore Bulawayo can still be considered for investment regardless of it having the longest simple payback period of 9.6 years.

Table 4. Summary of the 10 MW wind power systems in 28 locations in Zimbabwe.

Location	Capacity factor (%)	Simple payback (year)	Gross energy production (MWh)	Electricity exported to the grid (MWh)	Specific yield (kWh/m^2)	Net present value (NPV) US$	Annual life cycle savings (US$/year)	Energy Production cost (US $/MWh)	Net annual GHG reduction (tCO_2)	Barrels of crude oil not consumed (bbl)
Beitbridge	10.6	7.7	5,521	9,274	339	13,808,630	1,320,709	99.48	5,743	13,356
Bulawayo	8.6	9.6	4,488	7,539	275	7,876,793	753,366	129.46	4,669	10,858
Chegutu	10.5	7.8	5,465	9,180	335	13,484,663	1,289,724	100.82	5,685	13,221
Chikore	12.4	6.6	6,444	10,824	395	19,107,960	1,827,557	80.82	6,703	15,588
Chinhoyi	10.9	7.4	5,709	9,589	350	14,884,310	1,423,591	95,20	5,938	13,809
Chipinge	12.7	6.4	6,649	11,169	408	20,284,556	1,940,091	77.38	6,917	16,086
Chiredzi	11.5	7.1	5,984	10,051	367	16,462,593	1,574,544	89.41	6,224	14,474
Dorowa	11.2	7.3	5,843	9,815	359	15,656,501	1,497,446	92.30	6,078	14,135
Gokwe	11.2	7.2	5,856	9,837	359	15,730,752	1,504,548	92.03	6,092	14,167
Gwanda	9.1	9.1	4,731	7,947	290	9,270,660	886,681	121.24	4,921	11,444
Gweru	9.2	8.9	4,799	8,062	295	9,663,081	924,213	119,07	4,992	11,609
Gwetera	12.5	6.5	6,515	10,943	400	19,513,877	1,866,380	79.61	6,777	15,760
Harare	11.5	7.0	6,021	10,114	370	16,677,911	1,595,138	88.66	6,263	14,565
Kutsaga	10.7	7.6	5,569	9,355	342	14,084,821	1,347,125	98.35	5,793	13,472
Hwange	11.9	6.8	6,223	10,453	382	17,836,805	1,705,979	84.79	6,473	15,053
Inyati	9.3	8.9	4,838	8,126	297	9,882,536	945,203	117.89	5,032	11,702
Kamativi	12.0	6.8	6,267	10,527	385	18,090,563	1,730,249	83.97	6,519	15,160
Kariba	10.5	7.8	5,476	9,198	336	13,547,232	1,295,708	100.56	5,696	13,247
Karoi	10.8	7.5	5,654	9,497	347	14,569,096	1,393,436	96.43	5,881	13,677
Kwekwe	10.5	7.8	5,453	9,159	335	13,415,051	1,283,066	101.11	5,672	13,191
Marawa	11.0	7.4	5,752	9,662	353	15,132,706	1,447,349	94.25	5,983	13,914
Marondera	11.3	7.2	5,908	9,923	363	16,027,040	1,532,886	90.96	6,145	14,291
Masvingo	8.8	9.4	4,586	7,703	281	8,438,369	807,077	126.04	4,771	11,095
Mhangura	11.2	7.2	5,861	9,845	360	15,761,143	1,507,455	91.92	6,097	14,179
Mt Darwin	11.8	6.9	6,153	10,335	378	17,433,908	1,667,444	86.11	6,400	14,884
Mutare	12.4	6.5	6,460	10,850	396	19,195,907	1,835,968	80.55	6,719	15,626
Ruwa	12.8	6.3	6,658	11,184	409	20,338,006	1,945,203	77.23	6,926	16,107
Shurugwi	9.8	8.4	5,114	8,590	314	11,469,035	1,096,914	109.86	5,320	12,372

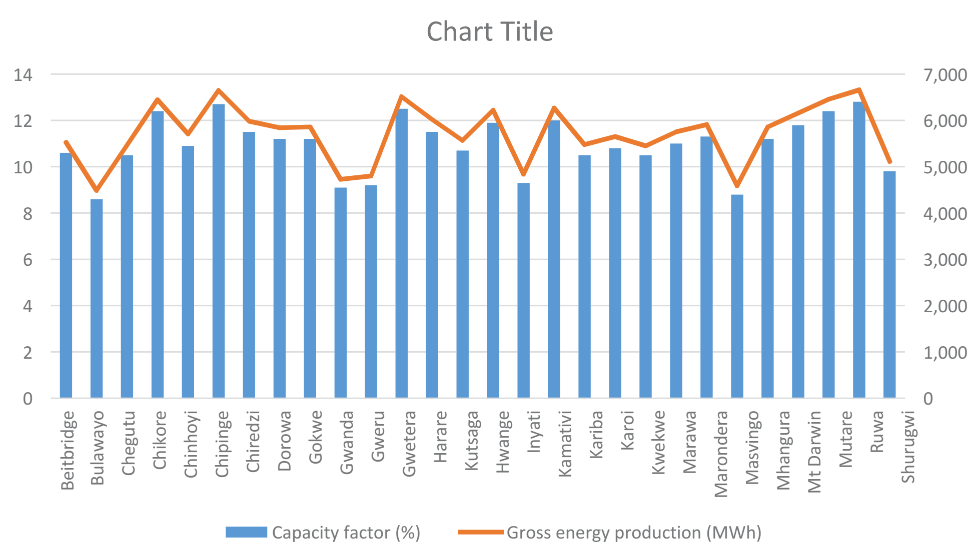

Figure 3. A relationship between the capacity factor and the gross energy production for the 28 locations in Zimbabwe.

The overall analysis of this present study outlines that all the sites under study are profitable and feasible for a 10 MW grid-connected wind farm construction. From a financial and techno-economic feasibility point of view, Ruwa location, having the highest profit, is highly preferable contrary to Bulawayo location which results in the minimum profit. Further emissions analysis shows that Ruwa has the largest net annual GHG emission reduction value of 6,926 tons of carbon dioxide whilst Bulawayo has a value of 4,669 tons of carbon dioxide, which is the minimum. Generally, all sites under study reduce net annual GHG emissions with an equivalent of more than 10,000 barrels of crude oil not consumed per year.

3.1. Rayleigh statistics

The Rayleigh distribution is the simplest probability distribution of wind speed which only requires the mean wind speed, v_m, to represent the wind resource. It is a representation of a Weibull distribution in which the dimensionless shape parameter, k = 2, making it a special case. The probability density function is given by (1) and the cumulative distribution function is given by (2) (Nage 2016).

$$f(v) = \frac{\pi v}{2 v_m} \exp\left[-\frac{\pi}{4}\left(\frac{v}{v_m}\right)^2\right] \quad (1)$$

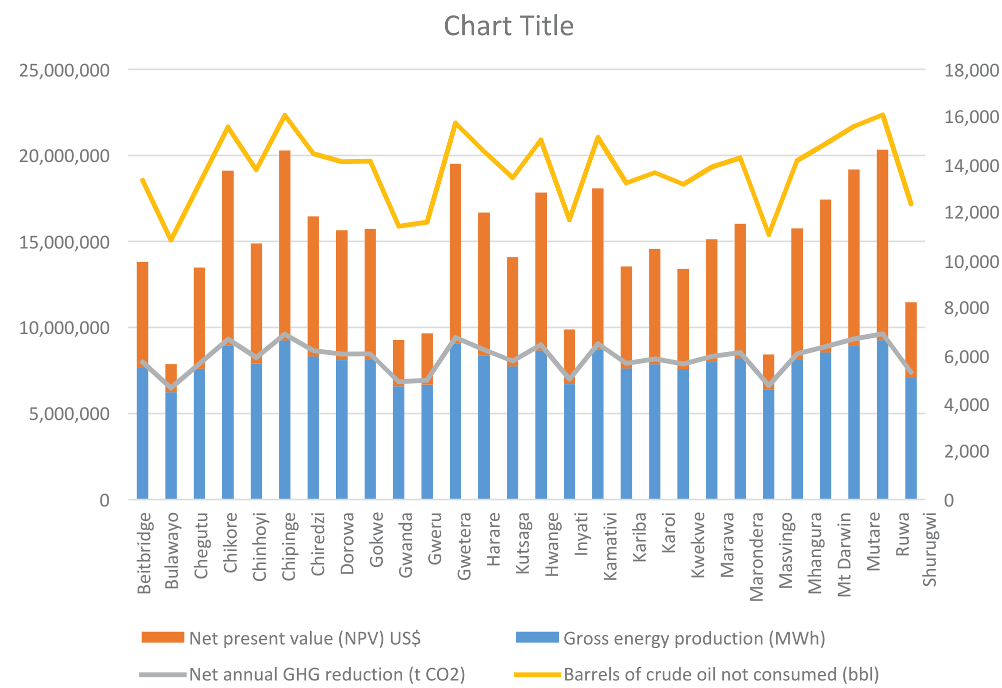

Figure 4. A relationship between the NPV, CO_2 reductions, the gross energy productions and the respective barrels of crude oil not consumed saved for the 28 locations in Zimbabwe.

$$F(v) = 1 - \exp\left[-\frac{\pi}{4}\left(\frac{v}{v_m}\right)^2\right] \tag{2}$$

Where v_m, is the mean wind speed and v, is the wind speed.

Further analyses on the wind resources for the validation of the obtained results were done by employing Rayleigh statics for the determination of the Probability Density Functions (PDFs) and the respective Cumulative Distribution Functions (CDFs) of the different locations. The outlined results are for the region with the maximum average wind speed (Ruwa), the modal average wind speed (Harare) and the site with the minimum average wind speed (Bulawayo) which also support that Ruwa is the most feasible location for the construction of a 10 MW wind farm as shown in Figures 5–7.

4. Conclusion

The techno-economic and environmental feasibility study, as well as the potential of a 10 MW grid-connected wind farm project, was evaluated for 28 locations in Zimbabwe. In this present study wind speed data from 1991–2010 were employed using Meteonorm V7 software into Typical Meteorological Year 2 (TMY2) format and RETScreen modeling software version 4.0 from Natural Resources Canada was then used for the analysis of the data.

The gross energy production in MWh of the Gamesa G132-5.0 wind turbine model ranged from 4,488 to 6,658 for the locations understudy with the minimum observed at Bulawayo and maximum at Ruwa. Furthermore, 7,500 to 11,200 MWh of energy can also be exported to the grid. The wind turbine model was selected because of its low cut-in speed of 1.5 m/s, a high cutout speed of 27 m/s and a rated power of 5,000 kW at a hub height of 95 m.

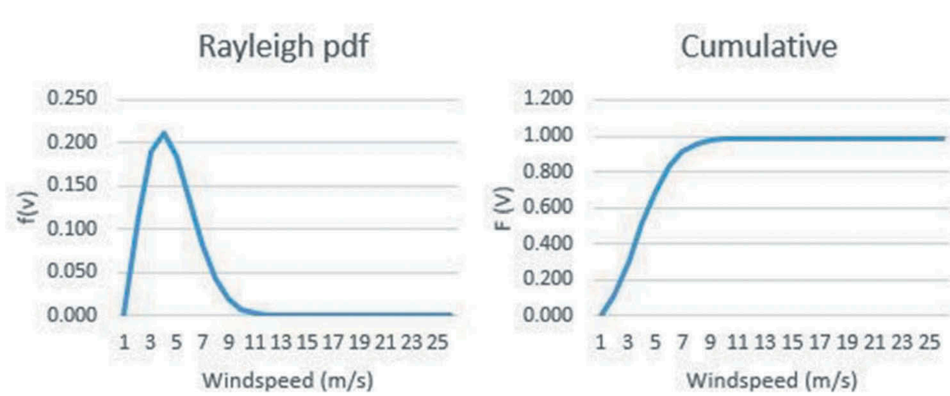

Figure 5. The PDF (f(v)) and CDF (F(v)) for Harare location.

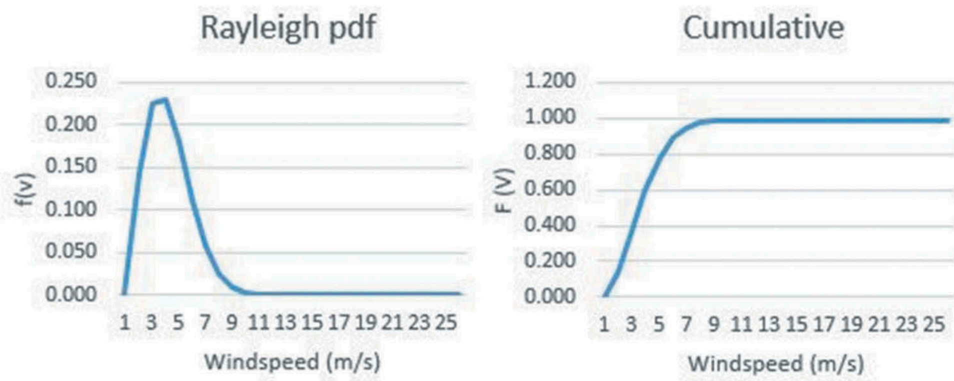

Figure 6. The PDF (f(v)) and CDF (F(v)) for Bulawayo location.

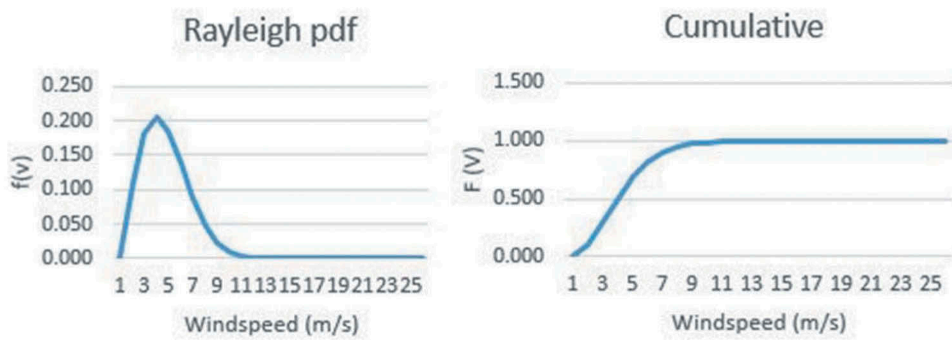

Figure 7. The PDF (f(v)) and CDF (F(v)) for Ruwa location.

For the investment of the proposed 10 MW wind farm, 242,811 m2/MW of land is required. The range of energy production cost is from a minimum of US$77.23/MWh observed at Ruwa to a maximum of US$129.46/MWh occurring at Bulawayo. Further analysis could be performed on the existing locations that NASA database could not provide their data for. A more accurate wind energy potential study can also be performed once the exact feed-in tariffs for wind energy is announced by the Zimbabwe Energy Regulatory Authority (ZERA).

ORCID

Remember Samu http://orcid.org/0000-0003-4531-4815

References

Adaramola, M., Agelin-Chaab, and S. S. Paul. Jan, 2014. Assessment of wind power generation along the coast of Ghana. *Energy Conversion and Management* 77:61–69. aecf. 2018. "Renewable Energy in Zimbabwe | AECF." 2018. https://www.aecfafrica.org/media_centre/blogs/renewable_energy_in_zimbabwe.

Africa-EU Renewable Energy Cooperation Program. 2015. https://www.africa-eu-renewables.org/market-information/zimbabwe/.

Aecf. 2018. "renewable energy in zimbabwe | aecf." 2018. https://www.aecfafrica.org/media_centre/blogs/renewable_energy_in_zimbabwe

Argin, M., and V. Yerci. 2017. Offshore wind power potential of the Black Sea Region in Turkey. *International Journal of Green Energy* 14 (10):811–18. doi:10.1080/15435075.2017.1331443.

Asumadu-Sarkodie, Samuel Asantewaa Owusu. 2016. "the potential and economic viability of wind farms in ghana." Energy Sources, Part A: Recovery, Utilization, and Environmental Effects 38: 695–701. https://doi.org/10.1080/15567036.2015

Aweo.org. 2015. "Area Used by Wind Power Facilities." http://www.aweo.org/windarea.html.

Ayodele, T. R., A. A. Jimoh, J. L. Munda, and J. T. Agee. 2013. A statistical analysis of wind distribution and wind power potential in the coastal region of South Africa. *International Journal of Green Energy* 10 (8):814–34. doi:10.1080/15435075.2012.727112.

Ayodele, T. R., A. A. Jimoh, J. L. Munda, and J. T. Agee. 2014. Viability and economic analysis of wind energy resource for power generation in Johannesburg, South Africa. *International Journal of Sustainable Energy* 33 (2):284–303. doi:10.1080/14786451.2012.762777.

Boudia, S. M., A. Benmansour, N. Ghellai, M. Benmedjahed, and M. A. Tabet Hellal. 2012. Monthly and seasonal assessment of wind energy potential in Mechria Region, Occidental Highlands of Algeria. *International Journal of Green Energy* 9 (3):243–55. doi:10.1080/15435075.2011.621482.

Cetinay, H., F. A. Kuipers, and A. Nezih Guven. 2017. Optimal siting and sizing of wind farms. *Renewable Energy* 101 (February):51–58. doi:10.1016/J.RENENE.2016.08.008.

Commin, A. N., and M. John. 2017. A case study of an isolated wind powered electricity system meeting 100% of local demand. *International Journal of Green Energy* 14 (13):1141–49. doi:10.1080/15435075.2017.1369417.

Electricity, Zimbabwe Company (ZETDC). 2017. "zimbabwe electricity and transmission distribution company (zetdc)." 2017. http://zetdc.co.zw/

EPA. 2016. EPA fact sheet social cost of carbon. https://www.epa.gov/sites/production/files/2016-12/documents/social_cost_of_carbon_fact_sheet.pdf.

Ersoz, S., H. Tahir Cetin Akinci, S. Nogay, and G. Dogan. 2013. Determination of wind energy potential in Kirklareli-Turkey. *International Journal of Green Energy* 10 (1):103–16. doi:10.1080/15435075.2011.641702.

Fant, C., B. Gunturu, and A. Schlosser. 2016. Characterizing wind power resource reliability in Southern Africa. *Applied Energy* 161 (January):565–73. doi:10.1016/j.apenergy.2015.08.069.

Gamesa. 2016. "Gamesa G132-5.0MW." http://www.indaba-southafrica.co.za/downloads/idn/2015/IDN2015-DAY-2-web.pdf.

Gamesa. 2016. "Gamesa Platform Catalogue." http://www.gamesacorp.com/recursos/doc/productos-servicios/aerogeneradores/nuevas-fichas/catalogo-plataformas-eng.pdf.

Genchi, S. A., A. J. Vitale, M. Cintia Piccolo, M. E. Gerardo, and Perillo. 2016. Wind energy potential assessment and techno-economic performance of wind turbines in coastal sites of Buenos Aires Province, Argentina. *International Journal of Green Energy* 13 (4):352–65. doi:10.1080/15435075.2014.952426.

Gualtieri, G. 2019. A comprehensive review on wind resource extrapolation models applied in wind energy. *Renewable and Sustainable Energy Reviews* 102 (March):215–33. doi:10.1016/J.RSER.2018.12.015.

Gualtieri, G., and G. Zappitelli. 2015. Investigating wind resource, turbulence intensity, and gust factor on mountain locations in Southern Italy. *International Journal of Green Energy* 12 (4):309–27. doi:10.1080/15435075.2013.840785.

Gugliani, G. K., A. Sarkar, S. Bhadani, and S. Mandal. 2019. A novel approach for accurate assessment of design wind speed for variable wind climate. *KSCE Journal of Civil Engineering* 23 (2):608–23. doi:10.1007/s12205-018-1431-6.

Gugliani, G. K., A. Sarkar, S. Mandal, and V. Agrawal. 2017. Location wise comparison of mixture distributions for assessment of wind power potential: A parametric study. *International Journal of Green Energy* 14 (9):737–53. doi:10.1080/15435075.2017.1327865.

Hedayat, Z., M. Bert Belmans, H. Ayatollahi, I. Wouters, and F. Descamps. 2017. Wind energy and natural ventilation potential of a wind catcher in Yazd – Iran (a long-term measurement). *International Journal of Green Energy* 14 (7):650–55. doi:10.1080/15435075.2017.1315807.

Henbest, S., Mills, L., Orlandi, I., Serhal, A., Pathania, R.. 2015. "Bloomberg New Energy Finance: Levelised Cost of Electricity." https://assets.publishing.service.gov.uk/media/57a08991e5274a31e0000154/61646_Levelised-Cost-of-Electricity-Peer-Review-Paper-FINAL.pdf

Himri, Y., A. Boudghene Stambouli, and B. Draoui. 2009. Prospects of wind farm development in Algeria. *Desalination* 239 (1–3):130–38. doi:10.1016/j.desal.2008.03.013.

Hove, Tawanda, Downmore Musademba. 2014. "mapping wind power density for zimbabwe: a suitable weibull-parameter calculation method." Journal of Energy in Southern Africa 25 (4). http://www.scielo.org.za/pdf/jesa/v25n4/04.pdf

Indaba daily news. 2015. "Only in Africa." https://www.indaba-southafrica.co.za/downloads/idn/2015/IDN2015-DAY-2-web.pdf

International Energy Agency. 2014. World energy outlook special report. Africa Energy Outlook.

Landry, M., Y. Ouedraogo, Y. Gagnon, and A. Ouedraogo. 2017. On the wind resource mapping of Burkina Faso. *International Journal of Green Energy* 14 (2):150–56. doi:10.1080/15435075.2016.1253571.

Makonese, T. 2016. Renewable energy in Zimbabwe. 2016 International Conference on the Domestic Use of Energy (DUE), 1–9. IEEE. doi:10.1109/DUE.2016.7466713.

Mandelli, S., J. Barbieri, L. Mattarolo, and E Colombo. 2014. sustainable energy in africa: a comprehensive data and policies review. *Renewable and Sustainable Energy Reviews* 2014 2014 37 (05):069. doi:10.1016/j.rser.2014.05.069.

Manomaiphiboon, K., C. P. Paton, T. Prabamroong, N. Rajpreeja, N. Assareh, and M. Siriwan. 2017. Wind energy potential analysis for Thailand: Uncertainty from wind maps and sensitivity to turbine technology. *International Journal of Green Energy* 14 (6):528–39. doi:10.1080/15435075.2017.1305963.

Mentis, Dimitrios, and Sebastian Hermann. 2015. "assessing the technical wind energy potential in africa a gis-based approach.". *Renewable Energy* 83 (November):110–25.

Mungwena. 2002. "The Distribution and Potential Utilizablity of Zimbabwe's Wind Energy Resource." Renewable Energy 26 (3): 363–77. https://doi.org/10.1016/S0960-1481(01)00141-0.

Nage, Girma. 2016. "analysis of wind speed distribution: comparative study of weibull to rayleigh probability density function; a case of two sites in ethiopia." american journal of modern energy 2 (3): 10–16. https://doi.org/10.11648/j. *Ajme* 20160203:11.

NREL.gov. 2015. Nrel.Gov. 2010 cost of wind energy review.

OpenEI. 2018. RETScreen clean energy project analysis software | Open energy information. https://openei.org/wiki/RETScreen_Clean_Energy_Project_Analysis_Software.

Oree, V., and A. Rajoo. 2015. The potential for wind-generated electricity in Mauritius. *International Journal of Green Energy* 12 (8):821–31. doi:10.1080/15435075.2014.888657.

RETScreen International. Clean Energy Project Analysis. n.d..

Sahin, B., and M. Bilgili. 2009. Wind characteristics and energy potential in Belen-Hatay, Turkey. *International Journal of Green Energy* 6 (2):157–72. doi:10.1080/15435070902784947.

Samu, R., M. Fahrioglu, and O. Taylan. 2016. Feasibility study of a grid connected hybrid PV- wind power plant in Gwanda, Zimbabwe. IEEE Honet Symposium, Turkish Republic of Northern Cyprus 122–26

Samu, R., and M. Fahrioglu. 2017. An analysis on the potential of solar photovoltaic power. *Energy Sources, Part B: Economics, Planning, and Policy* 12 (10):883–89. doi:10.1080/15567249.2017.1319437.

Santos, M., and G. Mario. 2019. Factors that influence the performance of wind farms. *Renewable Energy* 135 (May):643–51. doi:10.1016/J.RENENE.2018.12.033.

Sanusi, N., A. Zaharim, S. Mat, and K. Sopian. 2017. A weibull and finite mixture of the von mises distribution for wind analysis in Mersing, Malaysia. *International Journal of Green Energy* 14 (12):1057–62. doi:10.1080/15435075.2017.1357124.

Türk Toğrul, İ., and M. İ. Kizi. 2008. Determination of wind energy potential and wind speed data in Bishkek, Kyrgyzstan. *International Journal of Green Energy* 5 (3):157–73. doi:10.1080/15435070802106977.

Tawanda Hove, Luxmore Madiye, Downmore Musademba. 2014. "Mapping Wind Power Density for Zimbabwe: A Suitable Weibull-Parameter Calculation Method."Journal of Energy in Southern Africa 25 (4). http://www.scielo.org.za/pdf/jesa/v25n4/04.pdf

Ucar, A., and F. Balo. 2008. A seasonal analysis of wind turbine characteristics and wind power potential in Manisa, Turkey. *International Journal of Green Energy* 5 (6):466–79. doi:10.1080/15435070802498101.

Windenergyfoundation.org. 2015. "economics." http://www.windenergy-foundation.org/about-wind-energy/%0Aeconomics

World Bank. 2016. "The World Bank, Food, and Agriculture Organization." http://data.worldbank.org/indicator/AG.SRF.TOTL.K2.

Xydis, G. 2016. A wind resource assessment around large mountain Masses: The speed-up effect. *International Journal of Green Energy* 13 (6):616–23. doi:10.1080/15435075.2014.993763.

Yumak, H., T. Ucar, and S. Yayla. 2012. Wind energy potential on the coast of Lake Van. *International Journal of Green Energy* 9 (1):1–12. doi:10.1080/15435075.2011.617017.

ZERA Farmers INDABA Workshop 2015. 2015. http://www.indaba-southafrica.co.za/downloads/idn/2015/IDN2015-DAY-2-web.pdf.

Zimbabwe Central Bank Discount Rate. 2016. http://www.indexmundi.com/zimbabwe/central_bank_discount_rate.html.

Zimbabwe Inflation Rate. 2016. http://www.tradingeconomics.com/zimbabwe/inflation-cpi.

Zimbabwe Interest Rate. 2016. http://www.tradingeconomics.com/zimbabwe/interest-rate.